

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: 8th

Name of the Course: Mobile Application Development

Course Code: TCS822

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	What are the key differences between native, hybrid, and web apps? What are the advantages of developing a native app over a cross-platform app?	
	OR	
(b)	What are the best practices for mobile app UI/UX design? How do you decide which mobile platform (iOS, Android, or both) to develop for first?	
Q2	(10 Marks)	CO2
(a)	What is the role of an API in mobile app development? What are the advantages of using React Native for mobile app development?	
	OR	
(b)	What are the fundamental building blocks of a mobile application? How do front-end, back-end, and databases work together in a mobile app?	
Q3	(10 Marks)	CO2
(a)	What is the difference between Fragments and Activities in Android? What is the role of Navigation Controllers in mobile apps?	
	OR	
(b)	What is the purpose of a ViewModel in Android development? How do you manage the Fragment lifecycle ?	
Q4	(10 Marks)	CO1
(a)	What is the difference between dp, sp, px, and pt in Android development? What are the key components of an Android application ?	
	OR	
(b)	How do you create a new Android project in Android Studio ? What is the role of Gradle in Android development?	
Q5	(10 Marks)	CO1
(a)	What are the different types of layouts available in Android? What is the difference between ConstraintLayout, LinearLayout, and RelativeLayout ?	
	OR	
(b)	What are the different states of an Activity lifecycle ? How do you pass data between Activities using Intents ?	CO2

